

Module 9: Inheritance Genetics — Study Questions

1. Who is Gregor Mendel? What organism did he study?
2. What is the difference between a dominant allele and a recessive allele?
3. What is the difference between genotype and phenotype?
4. What is the difference between homozygous (AA, aa) and heterozygous (Aa)?
5. Cross $Tt \times Tt$ using a Punnett square. What is the genotypic ratio?
6. What is a test cross? What is it used for?
7. What phenotypic ratio do you expect from a dihybrid cross ($AaBb \times AaBb$)?
8. What is incomplete dominance? Give an example.
9. What is codominance? Give an example.
10. Explain the ABO blood type system. What are the possible genotypes for Type A blood?
11. Can two parents with Type A and Type B blood have a Type O child? Explain.
12. Why are X-linked recessive disorders more common in males?
13. A carrier mother ($X^B X^b$) and a normal father ($X^B Y$) have children. What fraction of sons will be colorblind?
14. Two unaffected parents have an affected child. Is the trait dominant or recessive?
15. What is a pedigree? What can it tell you?
16. What is a polygenic trait? Give one example.
17. How is the inheritance of human height different from the inheritance of pea plant height?

18. True or False: If you know someone's genotype, you always know their phenotype.
Explain.